

SLIP – 21

Section I: Data Structures [15 Marks]

Q1) Perform the following operations using static implementation of Queue i. Create a queue of n integers. ii. Delete the element from the queue and display updated queue.

Answer:

```
#include <stdio.h>
#define MAX 100

int q[MAX], front = 0, rear = -1;

void insert(int x)
{
    if(rear == MAX - 1)
        printf("Queue Overflow\n");
    else
        q[++rear] = x;
}

void delete()
{
    if(front > rear)
        printf("Queue Underflow\n");
    else
        printf("Deleted element = %d\n", q[front++]);
}

void display()
{
    int i;

    if(front > rear)
        printf("Queue is empty\n");
    else
    {
        for(i = front; i <= rear; i++)
            printf("%d ", q[i]);
        printf("\n");
    }
}

int main()
{
    int n, x, i;

    printf("Enter n: ");
    scanf("%d", &n);
```

```

        printf("Enter %d integers:\n", n);
for(i = 0; i < n; i++)
    {
        scanf("%d",
&x);        insert(x);
    }

    printf("Queue: ");
    display();

    delete();

    printf("Updated queue: "); display();

    return 0;
}

```

Section II: Database Management Systems [15 Marks]

Q 2) Consider the following database (Primary keys are underlined)

Bus (bno, capacity, depot_name)

Driver (dno, driver_name, license_no, address, age, salary)

The relationship is as follows: Bus and Driver are related with Many to Many relationships with the descriptive attribute Date_of_duty.

Q 2(A)

A. Create the above database in PostGreSQL and insert sufficient records. [5 marks]

Answer:

```

CREATE TABLE Bus(
    bno
    INT PRIMARY KEY,
    capacity INT,
    depot_name
    VARCHAR(50)
);

```

```

CREATE TABLE Driver(
    dno
    INT PRIMARY KEY,
    driver_name VARCHAR(50),
    license_no VARCHAR(20),
    address VARCHAR(100),
    age
    INT,

```

```

        salary NUMERIC(10,2)
    );

CREATE TABLE Bus_Driver(
    bno INT
REFERENCES Bus(bno),
    dno INT
REFERENCES Driver(dno),
    Date_of_duty DATE,
    PRIMARY KEY(bno, dno, Date_of_duty)
);

INSERT INTO Bus VALUES
(1,40,'Kothrud Depot'),
(2,15,'Swargate'), (3,30,'Kothrud
Depot');

INSERT INTO Driver VALUES
(1,'Swapnil','MH12-001','Pune',24,20000),
(2,'Suresh','MH12-002','Pune',40,25000),
(3,'Anil','MH12-003','Nashik',35,22000);

INSERT INTO Bus_Driver VALUES
(1,1,'2017-01-15'),
(3,2,'2009-02-12'),
(2,3,'2017-01-15');

```

Q 2(B)

B. Execute the following queries in PostGreSQL (any 3) [6 marks]

i. Find the number of buses having capacity more than 20.

Answer:

```

SELECT license_no FROM Driver WHERE
driver_name LIKE 'S%';

```

ii. Calculate the average salary of a driver.

Answer:

```

SELECT AVG(salary) FROM Driver;

```

iii. Find all details of driver named "Swapnil"

Answer:

```

SELECT * FROM Driver
WHERE driver_name = 'Swapnil';

```

Q 2(C)

C. Write a query to find the names of all drivers who were driving buses from “Kothrud Depot” on 12/2/2009. [4 marks]

Answer:

```
SELECT DISTINCT D.driver_name
FROM Driver D
JOIN Bus_Driver BD ON D.dno = BD.dno
JOIN Bus B ON B.bno = BD.bno WHERE
B.depot_name = 'Kothrud Depot' AND
BD.Date_of_duty = '2009-02-12';
```

Q 3) Viva [5 marks]